Ahmed Yousry Khalaf

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Profile

Dedicated Computer and Systems Engineering student with a strong academic foundation in AI and Data Science. Currently specializing in Data Engineering through the Digital Egypt Pioneers Initiative (DEPI), with hands-on learning in Python and SQL. Passionate about building efficient data pipelines and exploring the intersection of software engineering and big data. A fast learner and proactive problem-solver, actively seeking a summer internship to transition academic knowledge into professional practice and contribute to innovative AI projects.

Education

Minia University | *Bachelor of Engineering* (Sep 2023 – **Present**)

- **Major**: Computer and Systems Engineering.
- **Location**: Minia, Egypt.
- **Current GPA**: 3.7

Technical Skills

- **Programming Languages:** C, C++, Python, SQL.
- **Frameworks & Libraries:** NumPy, Pandas, Matplotlib, Seaborn.
- **Databases:** MySQL, SQL Server, MongoDB.
- **Tools:** MATLAB, Microsoft Office (Excel, Word, PowerPoint).
- **Concepts:** OOP, Data Structures, Algorithms.

Internship

AI & Data Science Trainee | Digital Egypt Pioneers Initiative (DEPI) (Nov 2025 – **Present**)

- Developing expertise in the end-to-end data pipeline, from Data Cleaning and Exploratory Data Analysis (EDA) to model deployment.
- Mastering SQL-based data modeling and advanced Machine Learning algorithms through rigorous practical training.
- Utilizing analytical thinking to solve real-world problems using industry-standard tools and methodologies.

Courses

Elements of AI Course | Tech Pathways MENA (Nov 2025)
(University of Helsinki & Google.org)

- Completed a comprehensive curriculum covering the fundamentals of Artificial Intelligence and its business applications.
- Gained deep insights into Machine Learning concepts, neural networks, and the ethical implications of AI technology.

Introduction to Arduino | TIEC & ITIDA (Creativa) (May 2025)

- Completed a hands-on workshop focused on embedded systems fundamentals and Arduino programming.
- Gained experience in interfacing sensors and actuators with microcontrollers to build automated prototypes.
- Developed practical skills in circuit design and troubleshooting electronic components.

Projects

Radar & Sonar Visualizer | Embedded Systems Project
Tech Stack: Arduino Uno, Ultrasonic Sensor, Servo Motor, Processing IDE.

- **System Design**: Developed a rotating sonar system using a servo motor to scan a 180° field, detecting objects via ultrasonic waves.
- **Data Visualization**: Integrated Arduino with Processing IDE to map real-time sensor data into a 2D radar interface.
- **Object Tracking**: Implemented algorithms to calculate distance and angle, providing a visual representation of the surroundings.

Simple Smart Dustbin | Arduino Project
Tech Stack: Arduino Uno, HC-SR04, SG90 Servo, C++.

- **Automation:** Built a touchless waste system using ultrasonic sensors to trigger automatic lid operation (threshold < 20cm).
- **Control Logic:** Programmed precise servo movements using PWM and conditional logic to ensure smooth and hygienic operation.

Languages

- Arabic (Native).
- English (Intermediate).